

Inventory Service Health Check

Q3 review of the fixture service — healthy, three bugs to fix

The service is stable: endpoints, tests, and seed data all hold

- 3 endpoints working
- 8 tests passing
- Fixture seeded

Three bugs block the release

- bulk_restock rebuilds the key list every iteration — $O(n^2)$
- empty-order bypass skips availability checks
- average_stock risks ZeroDivisionError on empty state

Bug summary: three fixes, all Important

Bug	Location	Severity
Off-by-one index pattern	bulk_restock	Important
Empty-order semantic bug	fulfill_by_skus	Important
ZeroDivisionError risk	average_stock	Important

Fix all three before shipping

- Refactor `bulk_restock` to iterate with `.items()`
- Validate empty orders before fulfillment
- Guard `average_stock` against an empty state

Next steps

- Land the three fixes in fixture v2
- Re-run the 8-test suite to confirm green